

SmartInvest: AI-Powered Investment Portfolio Analyzer

— Project Documentation

1. Executive Summary & Market Context

In the contemporary financial landscape, retail investors face an unprecedented paradox: access to more data than ever before, yet a diminishing ability to derive meaningful insight. This phenomenon, often termed "analysis paralysis," is exacerbated by the fragmentation of assets across multiple platforms—centralized brokerages, decentralized crypto wallets, and traditional physical assets. SmartInvest is engineered to serve as a singular, AI-driven intelligence layer that unifies this data, providing professional-grade analytics to the everyday investor. The target audience for SmartInvest encompasses three primary archetypes: the nearing-retirement investor seeking de-risking strategies, the active trader requiring real-time volatility assessments, and the long-term wealth builder focused on consistent growth tracking. By democratizing sophisticated portfolio management tools previously reserved for institutional clients, SmartInvest empowers users to align their complex holdings with specific, long-term financial objectives through the use of Generative AI.

2. System Architecture & Data Flow

SmartInvest utilizes a modular three-tier architecture designed for low-latency processing and high scalability. The system is partitioned into the following functional layers:

1. **Presentation Layer (Streamlit Frontend):** Acts as the user gateway, managing state for file uploads (CSV, PDF, Excel) and manual entry. It ensures a responsive UI that translates complex financial data into legible dashboards.
2. **Intelligence Layer (Groq & LLaMA 3.3-70B):** The analytical core. This layer receives sanitized portfolio data and performs qualitative risk assessments, sector correlation analysis, and rebalancing recommendations using high-speed inference.
3. **Data & Persistence Layer (Google Sheets API):** Provides a cloud-based storage solution. It maps local Python DataFrames to remote spreadsheet ranges, enabling historical performance tracking without the overhead of a dedicated SQL database.

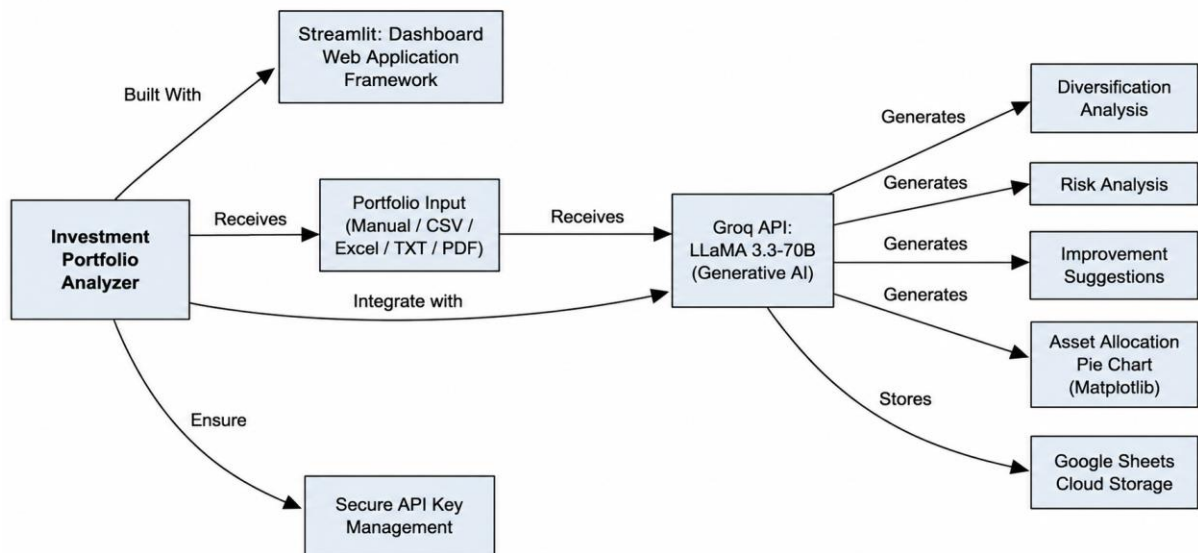
3. Technology Stack & Rationale

The selection of technologies for SmartInvest was driven by a need for rapid deployment, cost-efficiency, and industry-leading performance.

- **Streamlit vs. React/Flask:** Streamlit was chosen for its ability to build data-heavy applications entirely in Python, reducing the development cycle by eliminating the need for separate frontend/backend synchronization code.

- **Groq LLaMA 3.3-70B vs. OpenAI GPT-4:** Groq's LPU (Language Processing Unit) architecture offers significantly higher tokens-per-second, which is critical for real-time analysis of lengthy brokerage statements.
- **Google Sheets API vs. PostgreSQL:** For a user-centric tool, Google Sheets provides an intuitive "view-only" database that the user already knows how to use, ensuring data transparency and portability.
- **Scenario 1: Retirement Diversification Check**
A user nearing retirement uploads a PDF of their current brokerage statement. SmartInvest parses the document, identifies a heavy concentration in tech stocks, and provides an AI-generated suggestion for a safer, more balanced allocation to fixed-income assets.
- **Scenario 2: Active Trader Risk Assessment**
A crypto enthusiast manually enters their latest holdings across various altcoins. The system calculates a high composite risk score and generates a "Weakness Analysis" highlighting the volatility risks, recommending a shift toward more stable assets to protect gains.
- **Scenario 3: Portfolio Growth Tracking**
A long-term investor uploads their monthly CSV performance sheet. SmartInvest updates their profit/loss metrics and automatically pushes the new data to a cloud-based Google Sheet, allowing the investor to track their historical growth over time.

2. Technical Architecture



3. System Architecture & Logic

The application utilizes a modular three-tier architecture designed for scalability, security, and rapid data processing.

1. **Presentation Layer (Streamlit):** This layer serves as the user gateway. It handles multi-format file uploads and manual state management. It is designed with a responsive

layout to ensure that complex financial charts remain legible across all device types.

2. **Intelligence Layer (Groq & LLaMA 3.3):** The core "brain" of the system. It uses custom-engineered financial prompts to perform qualitative analysis. This layer identifies sector correlations and suggests rebalancing strategies that traditional algorithmic tools might miss.
3. **Data & Storage Layer (Google Sheets API & Matplotlib):** This layer ensures that insights are not ephemeral. It maps structured data to cloud-based spreadsheets for historical tracking and generates high-resolution visualizations for immediate feedback.

4. Tech Stack & Technical Pre-requisites

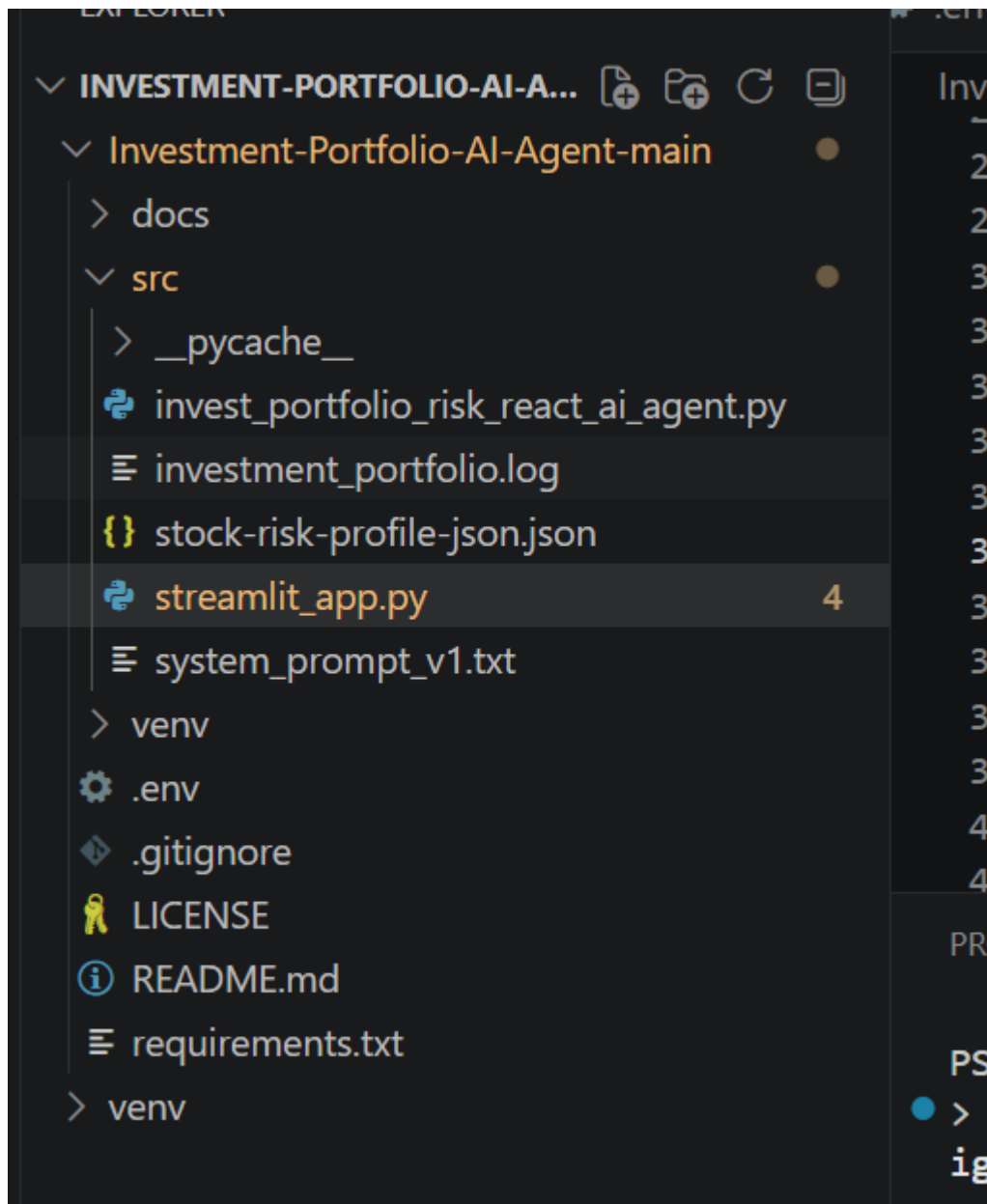
- **Streamlit Framework:** Chosen for its ability to turn Python scripts into interactive web apps without the overhead of traditional frontend frameworks.
- **Groq LLaMA 3.3-70B:** Utilized for its industry-leading inference speed and high context window, allowing for the processing of large brokerage statements without truncation.
- **Google Sheets API:** Provides a low-friction database alternative, allowing users to view their data in a familiar environment while ensuring cloud persistence.
- **Financial Libraries:** Pandas for vectorized calculations and Matplotlib for deterministic visualization of asset spreads.

- **Security Management:** Strict adherence to environment variable masking via .env files to protect sensitive API keys.

4. Project Milestones & Granular Implementation

Milestone 1: Environment Setup & Security Baseline

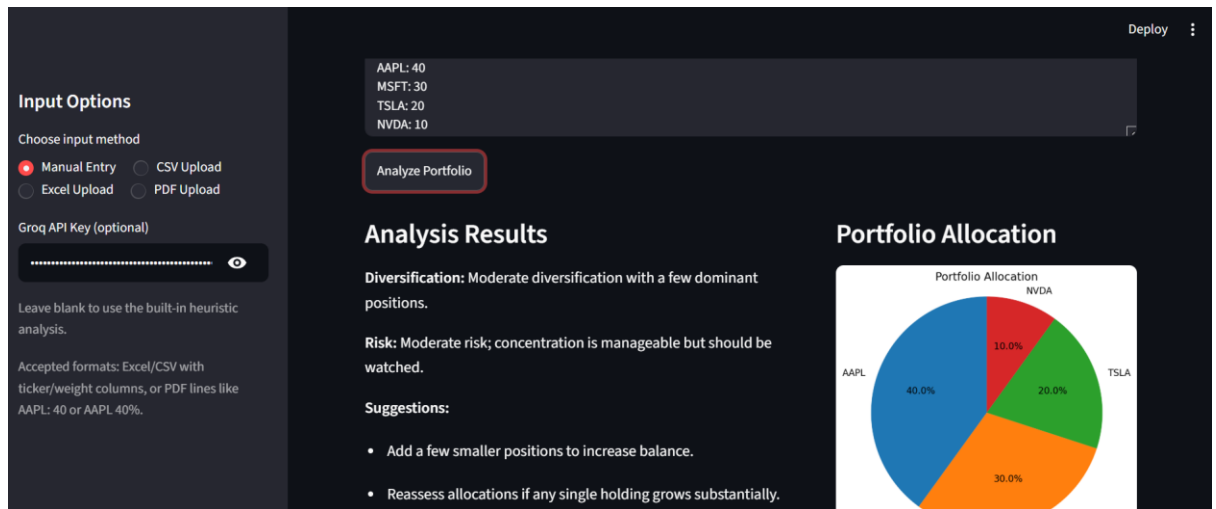
- **Activity 1.1:** Configuration of the Groq SDK and establishing a secure handshake with the LLaMA 3.3-70B model.
- **Activity 1.2:** Google Cloud Service Account provisioning with OAuth 2.0 scoping limited to necessary spreadsheet permissions.
- **Advanced Activity 1.3:** Security Audit — Implementing a pre-commit hook to prevent sensitive keys from ever entering version control.



- - **Project Structure**

Milestone 2: Multi-Channel Data Intake & Sanitization

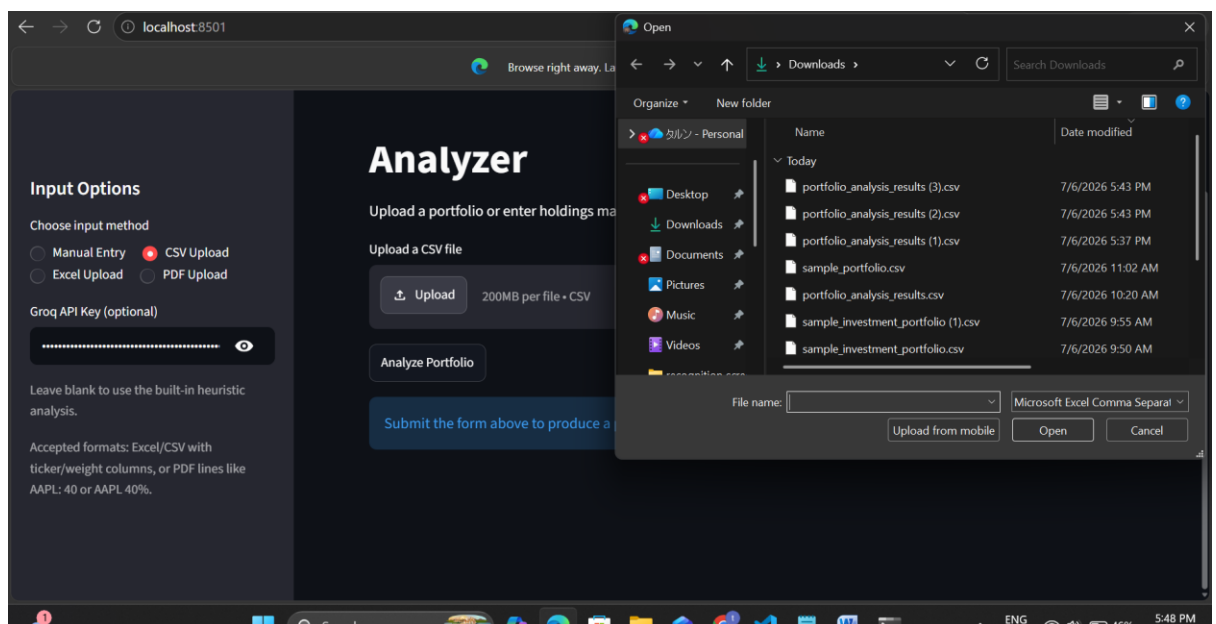
- **Activity 2.1:** Developing the ingestion pipeline for CSV and Excel files using the Pandas engine for robust schema validation.
- **Advanced Activity 2.2:** AI-Assisted Document Parsing — Using regex-based fallbacks to handle malformed PDF text extracted via PyPDF2.



○ Predicting Manual Entry

Milestone 3: Financial Analytics & Visual Engine

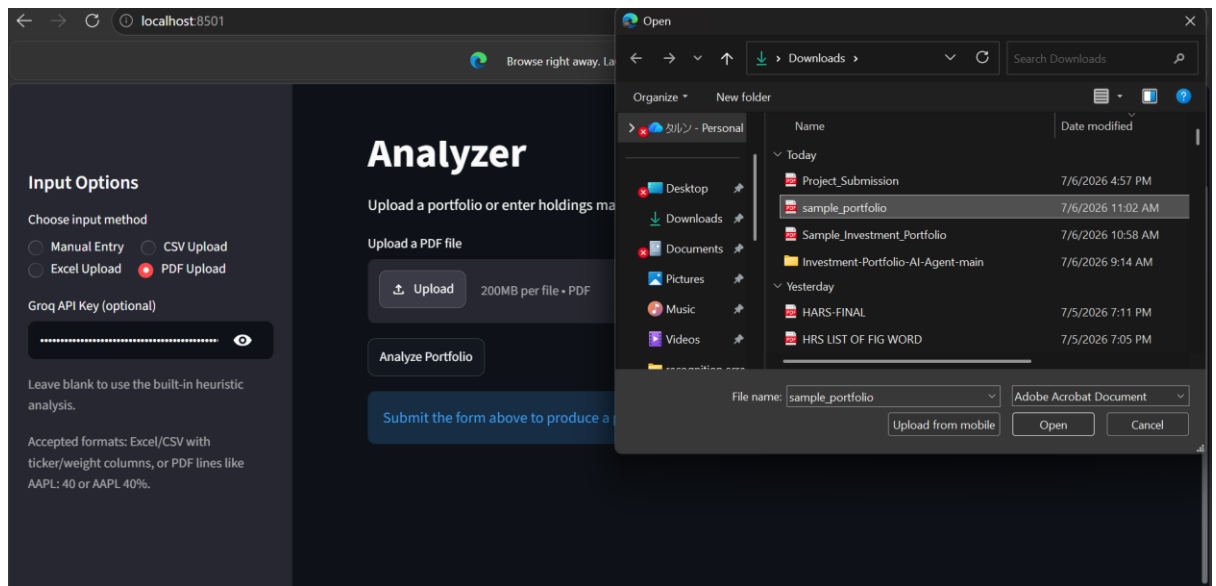
- **Activity 3.1:** Core Logic Implementation — Calculating weighted average returns and volatility benchmarks across asset classes.
- **Advanced Activity 3.2:** Interactive Visualization — Enhancing Matplotlib charts with dynamic legends and clear, print-ready asset labels.



- **Predicting Through Sample CSV file**

Milestone 4: Advanced Prompt Engineering

- **Activity 4.1:** System Instruction Design — Forcing the LLM to adopt a "Senior Investment Strategist" persona for all qualitative outputs.
- **Advanced Activity 4.2:** Output Structuring — Implementing Few-Shot prompting to ensure AI responses always conform to a parseable JSON-like format for the dashboard.



Predicting Through Sample PDF file

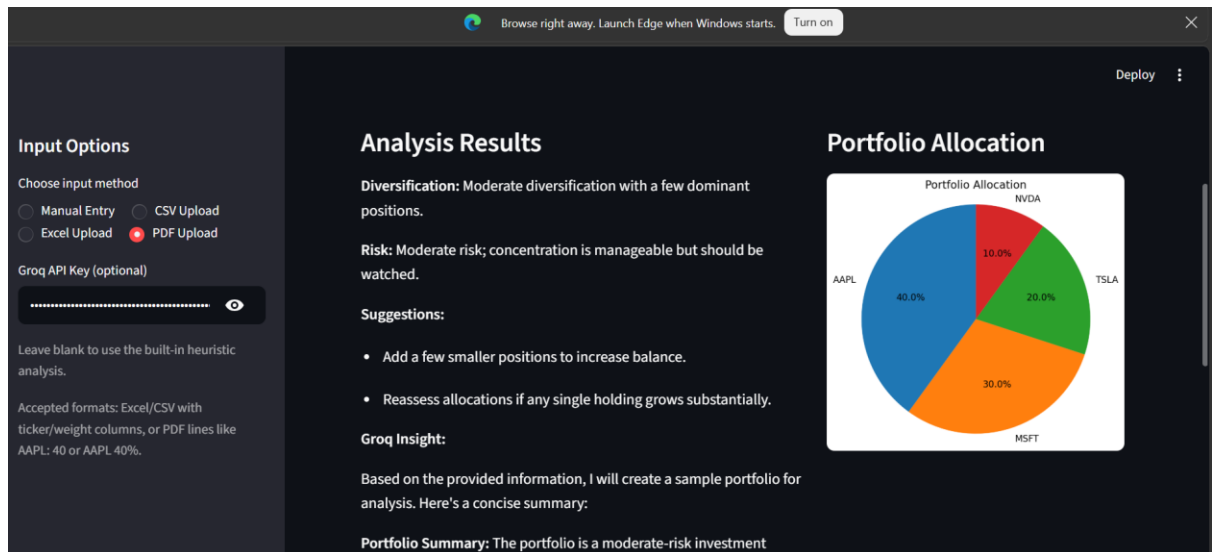
Milestone 5: Cloud Synchronization & Reliability

- **Activity 5.1:** Bi-directional mapping between Python Dataframes and Google Sheets ranges using Gspread.
- **Advanced Activity 5.2:** Error Handling & Retries — Implementing exponential backoff for API calls to prevent service interruptions during high traffic.

Milestone 6: UI/UX Finalization & Deployment

- **Activity 6.1:** Deployment — Finalizing the Streamlit Cloud or local Docker containerization for public availability.

- **Advanced Activity 6.2:** User Feedback Loop — Integrating an inline "Report Error" feature that logs hallucination events to help refine future prompts.



Predicting Results

5. Advanced Prompt Construction Strategies

The effectiveness of SmartInvest relies on high-fidelity prompt engineering. To ensure financial accuracy and avoid "AI hallucinations," we utilize a multi-step prompting strategy:

Qualitative Analysis Prompt Architecture

The system prompt is structured as follows:

```
def build_investment_prompt(portfolio_data):
    return f"""
    You are a professional Financial Risk Consultant.
    Analyze this raw portfolio data: {portfolio_data}
    Tasks:
    1. Quantify diversification on a 1-10 scale based on sector spread.
    2. Flag specific concentration risks (e.g., >30% in one asset).
```

3. Suggest rebalancing to safer assets based on current market trends.

Ensure the tone is professional, objective, and data-driven.

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□6. UI/UX Philosophy & Deployment Strategy

Design Philosophy: Clarity Over Complexity

The dashboard is designed using a "Top-Down" visual hierarchy. Users are first greeted with executive metrics—Total Value and Profit/Loss—before being presented with complex visualizations. This prevents information overload and focuses the user on the most critical KPIs immediately.

Cloud Synchronization & Deployment

The platform is architected for cloud-native deployment. By utilizing Streamlit Cloud, the application benefits from automatic SSL termination and seamless GitHub integration. For enterprise scaling, the application is containerized using Docker, allowing for deployment on AWS ECS or Google Cloud Run, ensuring that SmartInvest can handle thousands of concurrent analysis requests.

7. Conclusion

SmartInvest represents a paradigm shift in personal finance tools, moving from passive tracking to active, AI-driven advisory. By leveraging the low-latency capabilities of Groq and the ubiquity of Google Sheets, we have created a platform that is both

technologically advanced and practically accessible. Future iterations will focus on real-time API integrations with global brokerages and the addition of multi-currency tax optimization reports, further establishing SmartInvest as the premier open-source portfolio analyzer.

Document Prepared By:

Date:

7. User Guide: Dashboard Interpretation

SmartInvest follows a "Clarity Over Complexity" philosophy. The dashboard is divided into three primary zones:

- **Executive Metrics:** Displays Total Portfolio Value, Net Profit/Loss, and the AI-calculated "Health Score."
- **Visual Engine:** Interactive donut charts for sector allocation and line graphs for historical growth trends.
- **AI Advisory Panel:** A detailed report generated by the LLaMA 3.3 model, providing specific "Strong Signals" and "Immediate Risks" with actionable rebalancing steps.

8. Future Roadmap

The evolution of SmartInvest will focus on increasing automation and analytical depth:

- **Direct Brokerage Integration:** Moving from manual uploads to real-time Plaid or Yodlee API connections.
- **Tax-Loss Harvesting Engine:** Implementing logic to identify specific tax-saving opportunities during market downturns.
- **Multi-Currency Support:** Automated conversion for international assets across global exchanges.